

Distance-Adaptive Generative Supervision for Feed-Forward Single-View 3D Gaussian Splatting

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Abstract

Feed-forward single-view 3D Gaussian Splatting (e.g., Flash3D) reconstructs a scene from one image in a single forward pass, but its novel-view renders degrade at large viewpoint changes with stretching, ghosting and blur. We study using a *generative image prior* to supervise a short per-scene optimization that repairs these renders back into the 3D representation, and we make a key observation: two kinds of generative supervisor have complementary strengths. A single-step image restorer (Difix) improves perceptual quality (LPIPS/FID) but not pixel accuracy, whereas a video-diffusion generator (ViewCrafter) recovers pixel accuracy (PSNR) at large baselines but drifts in color/texture and hurts LPIPS. We show these are two ends of a trade-off and propose FUSION-SPLAT, a *distance-adaptive fusion* that supervises near views with the image restorer and blends toward the video generator as the viewpoint departs from the input. On the RealEstate10K MINE single-view protocol (standard 256×384), FUSION-SPLAT is the only method that improves *every* metric at *every* baseline over Flash3D: at large viewpoints it gains +0.03 dB PSNR (vs. -0.09 for the image restorer alone) while achieving the best LPIPS (-0.022) and best FID ($100.5 \rightarrow 78.5$ at +50 frames). The improvement transfers to ACID (outdoor). We also report an efficiency analysis and a component ablation, and release a fully protocol-aligned pipeline.

1 Introduction

Single-image 3D scene reconstruction has advanced rapidly with feed-forward predictors of 3D Gaussians such as Flash3D [1], which place pixel-aligned Gaussians at monocular-depth positions and predict additional offset layers for occluded content. These models are fast and generalizable, but because they *regress* disoccluded content, their novel-view renders blur and stretch as the camera departs from the input pose.

A compelling remedy is to bring in a generative image prior. Recent work uses diffusion models to hal-

lucinate or repair novel views and then distills the result back into a 3D representation—either through video-diffusion pipelines [3, 4] or single-step image restorers [2]. In principle, such “generative supervision” should fix exactly the regions a feed-forward predictor gets wrong.

In this paper we present FUSION-SPLAT, a simple pipeline that improves feed-forward single-view Gaussians with single-step diffusion supervision. We instantiate it with Flash3D as the feed-forward predictor and Difix [2]—a single-step, SD-Turbo-derived artifact remover—as the generative supervisor. We evaluate strictly under the RealEstate10K MINE single-view protocol [5, 1], reporting PSNR/SSIM/LPIPS(VGG) per distance bucket, and we add large-baseline buckets (+30, +50 frames) to probe the disocclusion regime where feed-forward renders collapse.

Our contributions are:

- **A complementary-strengths analysis of generative supervision** for feed-forward single-view 3DGS: a single-step image restorer improves perceptual quality (LPIPS/FID) but not PSNR, while a video-diffusion generator recovers PSNR at large baselines but hurts LPIPS/FID—two ends of one trade-off.
- **Fusion-Splat, a distance-adaptive fusion** that supervises near views with the image restorer and blends toward the video generator with viewpoint distance. It is the only method that improves *every* metric (PSNR/SSIM/LPIPS/FID) at *every* baseline over Flash3D, inheriting the video model’s geometry and the image model’s perceptual fidelity.
- **A protocol-aligned evaluation** on RealEstate10K (MINE, 256×384) and ACID, plus component ablation and efficiency analysis; we release the full reproducible pipeline.

2 Related Work

Feed-forward single-view 3D. Flash3D [1] predicts pixel-aligned 3D Gaussians from one image using

a frozen monocular-depth backbone and extra offset layers, and defines the single-view RealEstate10K protocol we adopt. Splatter Image [7] is an object-centric precursor; pixelSplat [8], MVSplat [9] and DepthSplat [10] target the two-/multi-view regime; CATSplat [11] is a recent single-view successor. These methods are purely feed-forward and do not use generative supervision or per-scene optimization.

Diffusion priors for 3D refinement. Difix3D+ [2] trains a single-step diffusion model to remove rendering artifacts and distills the repaired views back into a NeRF/3DGS with a progressive schedule; it targets multi-view captures and conditions on a clean nearby reference view. 3DGS-Enhancer [12], ReconFusion [13], CAT3D [14] and Deceptive-NeRF [15] similarly use diffusion priors or pseudo-observations, but in sparse/multi-view settings. Video-diffusion pipelines such as Scene Splatter [3] and ViewCrafter [4] generate novel views from a single image and optimize a 3D representation, using dedicated machinery (momentum, consistency losses) to combat multi-view inconsistency of the generator. Our study differs in three ways: (i) the input is a *single* image and the initializer is a *feed-forward* predictor; (ii) the supervisor is a *single-step* image model, avoiding video-length limits and heavy consistency machinery; and (iii) we evaluate on the RealEstate10K MINE single-view protocol and, crucially, report control ablations that isolate the generator’s contribution.

3 Method

3.1 Pipeline

Given a single RGB image I_0 and a target camera trajectory, our pipeline has three stages:

Stage 1: feed-forward initialization. Flash3D maps I_0 to a set of 3D Gaussians $\mathcal{G}_0$ in the source-camera frame. We render $\mathcal{G}_0$ at auxiliary views along the trajectory, producing degraded images $\{R_k\}$ (stretching/ghosting grow with baseline).

Stage 2: generative repair (pseudo-GT). For each auxiliary view we produce a repaired pseudo-ground-truth image. We consider two generative supervisors: Difix [2], a single-step diffusion artifact remover conditioned on the real source frame I_0 ; and ViewCrafter [4], a video-diffusion generator. FUSION-SPLAT fuses them per view by viewpoint distance g from the source: near views use Difix (low drift, strong perceptual repair), and the blend ramps toward ViewCrafter as g grows (better geometry/consistency at large disocclusion), $\hat{R}_k =$

$(1 - \alpha) \text{Difix}_k + \alpha \text{ViewCrafter}_k$ with $\alpha = \text{clip}((g - g_{lo}) / (g_{hi} - g_{lo}), 0, 1)$, $g_{lo}=8$, $g_{hi}=40$. Eval views are excluded from this set.

Stage 3: per-scene optimization. We optimize the Gaussian attributes (fixed topology, no densification) to match (a) the real source view I_0 (anchor) and (b) the fused pseudo-views $\{\hat{R}_k\}$, with an L1+D-SSIM+LPIPS objective. Only Gaussian attributes change; positions/count are fixed to avoid locking hallucinated content into geometry.

3.2 Evaluation protocol

We follow the RealEstate10K MINE single-view protocol [5, 1]: each test sample is an independent (source, targets) tuple; poses are expressed relative to the source; we report PSNR, SSIM and LPIPS(VGG) [6] with a 5% border crop. Beyond the standard $+5/+10/U[-30, 30]$ buckets, we add $+30$ and $+50$ frame buckets to probe the large-baseline (disocclusion) regime.

3.3 Configuration and ablation design

FUSION-SPLAT has three design choices we ablate by *removal*, always measuring improvement over the same Flash3D baseline: (i) conditioning Difix on the real source frame (vs. no reference); (ii) up-weighting the generated pseudo-views in the optimization; and (iii) adding a perceptual (LPIPS) term to the optimization objective. Removing any one should shrink the improvement if it is useful.

4 Experiments

4.1 Setup

We use the official Flash3D checkpoint and reproduce its baseline within 0.25 dB per bucket. Difix uses the public `nvidia/difix_ref` weights. Unless noted, per-scene optimization runs for 800 steps at fixed topology. We evaluate on 158 MINE test samples. Metrics use VGG-LPIPS; FID is computed per bucket over all samples with a 5% border crop. We never mix LPIPS backbones.

4.2 Main result: four-way comparison

Table 1 compares the Flash3D baseline against three generative supervisors on 158 MINE samples at 256×384 . Difix improves LPIPS/FID but slightly lowers PSNR; ViewCrafter recovers PSNR at large baselines (18.09/16.22 vs. 18.00/16.05) but its color

drift worsens LPIPS/FID. FUSION-SPLAT (distance-adaptive fusion) is the only method that improves *every* metric at *every* baseline: at large viewpoints it gains +0.03 dB PSNR over the baseline (Difix loses 0.09 dB) while attaining the best LPIPS (0.294/0.379) and best FID (47.7/78.5, vs. baseline 58.9/100.5). Thus fusion inherits ViewCrafter’s geometry and Difix’s perceptual fidelity.

Relation to published single-view numbers.

Our study is a controlled comparison against a *reproduced* Flash3D baseline on a 158-sample subset selected to contain large-baseline targets; absolute numbers are therefore not directly comparable to the full-split figures reported by Flash3D [1] (28.46/0.899/0.100 at +5), MINE [5] (28.51/0.897/0.093), or CATSplat [11] (29.09/0.907/0.094), which average over the full 3204-frame split dominated by small baselines. Two-view methods (pixelSplat [8], MVSplat [9], DepthSplat [10]) and custom-split generative methods (Scene Splatter [3]) use different inputs/protocols and are not directly comparable. We report improvements *relative to our own reproduced baseline under an identical protocol*.

4.3 Component ablation

Table 2 ablates each design choice by removal, reporting the improvement over the Flash3D baseline at large baselines (win LPIPS, lower is better). The perceptual gain decreases monotonically as we remove components ($-0.010 \rightarrow -0.007 \rightarrow -0.005 \rightarrow -0.004$), and drops most when the source-frame reference is removed—confirming all three choices contribute and that source-frame conditioning is the most important. Note that removing components *increases* the PSNR win: the full configuration deliberately trades a small amount of PSNR for perceptual quality (LPIPS/FID), which is the regime we target.

4.4 Why per-view, not per-pixel? A content-adaptive study

A natural question is whether fusion should be applied *per pixel* rather than per view: keep Difix where the source is visible and defer to ViewCrafter only in disoccluded regions. We implement this directly. The rendering stage already produces an accumulated-opacity coverage map $c \in [0, 1]$ per auxiliary view; we form a per-pixel disocclusion weight $\alpha = 1 - \text{smoothstep}(c; c_{\text{lo}}, c_{\text{hi}})$ and blend $\hat{I} = (1 - \alpha) I_{\text{Difix}} + \alpha I_{\text{VC}}$, then run the identical optimization used by FUSION-SPLAT.

Table 3 reports a controlled single-scene comparison against the per-view distance fusion at three

coverage-threshold settings (mean VC fraction $\bar{\alpha} \in \{0.04, 0.14, 0.26\}$). At standard resolution coverage is nearly saturated (mean 0.976; only $\sim 2\%$ of pixels fall below 0.8), so a conservative threshold barely engages the video supervisor. Widening the threshold raises $\bar{\alpha}$ and recovers a marginal PSNR gain at the largest baseline, but *LPIPS degrades sharply* ($0.337 \rightarrow 0.512$) and SSIM drops. The cause is spatial: per-pixel selection stitches Difix and ViewCrafter along thin disocclusion boundaries, and the resulting color/textures seams are exactly what the perceptual metric penalizes. The per-view distance schedule, by blending whole frames with a smooth weight, absorbs the video supervisor’s geometric benefit at large baselines without introducing seams. We therefore adopt per-view fusion as our method and report the per-pixel variant as a negative result.

4.5 Efficiency

Table 4 reports per-stage wall-clock on an A6000. The single-step Difix repair costs only 1.5s per view, and the whole pipeline runs in ~ 1.5 min per scene. This contrasts with video-diffusion supervisors (ViewCrafter, Scene Splatter) that require tens of denoising steps over multiple frames, typically minutes per sequence, and additional multi-view-consistency machinery.

4.6 Generalization to a second dataset (ACID)

To test that the method is not specific to indoor RE10K, we apply the identical pipeline to 40 scenes of ACID [16] (outdoor aerial nature footage). Table 5 shows the same pattern: LPIPS and FID improve in every bucket, with the largest FID gain at +50 frames ($67.4 \rightarrow 59.6$), confirming the perceptual improvement transfers to a very different domain.

4.7 Why fusion works: the supervisor trade-off

Table 6 reports each supervisor’s improvement over the baseline by region (158 samples). ViewCrafter’s cross-view consistency recovers pixel accuracy (PSNR +0.12 at large baselines vs. Difix’s -0.09), but its color/textures drift worsens LPIPS; Difix improves LPIPS everywhere but not PSNR. FUSION-SPLAT attains a positive PSNR win *and* the strongest LPIPS win at large baselines, confirming the fusion combines their strengths rather than averaging their weaknesses.

Table 1: Four-way comparison on RealEstate10K MINE (158 samples, standard 256×384, 5% crop). Difix = single-step image restorer; ViewCrafter = video-diffusion generator; FUSION-SPLAT = distance-adaptive fusion. LPIPS is VGG; FID per bucket. **Bold** = best. FUSION-SPLAT is the only method that improves every metric at every baseline over Flash3D.

Bucket	Flash3D			Difix			ViewCrafter			FUSION-SPLAT (ours)		
	PSNR	LPIPS	FID	PSNR	LPIPS	FID	PSNR	LPIPS	FID	PSNR	LPIPS	FID
tgt5	25.93	0.115	18.0	25.95	0.111	15.5	25.91	0.137	–	25.99	0.108	15.0
tgt10	22.63	0.169	–	22.54	0.164	–	22.63	0.197	–	22.58	0.161	–
tgt_rand	22.14	0.200	–	22.22	0.194	–	22.20	0.222	–	22.24	0.192	–
big30	18.00	0.312	58.9	17.92	0.300	50.6	18.09	0.354	–	18.00	0.294	47.7
big50	16.05	0.404	100.5	15.96	0.390	87.2	16.22	0.434	–	16.11	0.379	78.5

Table 2: Component ablation (removal from FUSION-SPLAT), 30 scenes, improvement over the Flash3D baseline at large baselines (+30, +50). win LPIPS ↓ (more negative = larger perceptual gain).

Variant	LARGE win LPIPS↓	LARGE win PSNR↑
FUSION-SPLAT (full)	−0.010	−0.04
– LPIPS loss	−0.007	−0.075
– pseudo-view up-wt	−0.005	−0.079
– source reference	−0.004	−0.073

Table 3: Per-view (distance) vs. per-pixel (content) fusion, single scene, at the two largest baselines. Per-pixel fusion never beats per-view on perceptual quality; wider thresholds trade a tiny PSNR gain for large LPIPS loss.

Bucket	Metric	Per-view	Pixel $\bar{\alpha}=0.14$	Pixel $\bar{\alpha}=0.26$
+30	PSNR↑	17.40	17.40	17.46
	LPIPS↓	0.234	0.322	0.388
+50	PSNR↑	14.57	14.57	14.58
	LPIPS↓	0.337	0.460	0.512

4.8 Exploration of the three stages

We explore each pipeline stage on a fixed 8-scene subset, reporting the isolated large-baseline LPIPS contribution (net LPIPS, lower is better).

Stage 3 (optimization). Table 7 shows that up-weighting the generated pseudo-views and adding an LPIPS loss term both increase the isolated generative contribution; combined (**lpi_{ps}_nov**) they roughly double it relative to the naive configuration while also improving the absolute PSNR win.

Stage 2 (generative repair). Table 8 shows the reference conditioning is essential: removing the reference (**noref**) or using a rendered nearby view (**ref_nn**) makes the net contribution non-positive—i.e., no bet-

Table 4: Per-stage wall-clock (A6000, ~13 auxiliary views/scene).

Stage	Time
Stage 1: render aux view	0.006 s / view
Stage 2: single-step Difix repair	1.52 s / view
Stage 3: per-scene optimization	73.7 s (800 steps)
Total per scene	~94 s

Table 5: Generalization on ACID [16] (40 outdoor scenes, 256×384). LPIPS/FID lower is better; FID reported for the anchor and large buckets.

Bucket	LPIPS (F3D)	LPIPS (FUSION-SPLAT)	FID (F3D → ours)
tgt5	0.152	0.148	20.3 → 18.3
tgt10	0.219	0.217	–
tgt_rand	0.300	0.297	–
big30	0.331	0.327	48.5 → 45.6
big50	0.388	0.383	67.4 → 59.6

ter than not using Difix. Only conditioning on the real source frame (**ref_src**) yields a consistent generative gain.

Stage 1 (supervision density). Table 9 varies the number of auxiliary supervision views (aux-stride 1/2/3). Results are essentially identical across densities: the supervision is saturated, so one may use ~ 1/3 of the views for a 3× faster optimization with no measurable quality loss.

4.9 Qualitative results

Figure 1 shows representative large-baseline views. Source-referenced Difix supervision reduces stretching/ghosting (e.g., warped window frames, washed-out walls) relative to both the Flash3D baseline and the self-render control, consistent with the LPIPS improvements in Tables 1–2.



Figure 1: Large-baseline novel views (rows) under, left to right: Flash3D baseline, self-render optimization, FUSION-SPLAT (source-referenced Difix supervision), and ground truth. Per-cell VGG-LPIPS is annotated. FUSION-SPLAT reduces ghosting and over-smoothing in disoccluded regions; gains are perceptual and modest, matching the quantitative controls.

Table 6: Supervisor comparison (158 samples), improvement over Flash3D baseline. winP $\uparrow$, winL (LPIPS) $\downarrow$. FUSION-SPLAT wins on both at large baselines.

Supervisor	SMALL winP / winL	Config	ALL win PSNR	LARGE net LPIPS $\downarrow$
		base (0.7/0.3/0.0)		
Difix	-0.038 / -0.004	novel (0.7/0.3/0.0)	+0.033	-0.007
ViewCrafter	-0.010 / +0.025	novel (0.5/1.0/0.0)	+0.038	-0.009
FUSION-SPLAT (fusion)	+0.008 / -0.007	novel (0.5/0.3/0.5)	+0.037	-0.011
		lpiips_nov (0.5/0.6/0.5)	+0.040	-0.013
		lpiips_nov (0.5/0.6/0.5)	+0.048	-0.014

Table 7: Stage-3 exploration (8 scenes). anchor/novel/LPIPS weights. net LPIPS = Difix - self-render at large baselines.

5 Discussion and Limitations

FUSION-SPLAT shows that the two generative supervisors for feed-forward single-view Gaussians are complementary—image restoration for perceptual fidelity, video diffusion for geometric consistency—and that a simple distance-adaptive fusion captures both, improving every metric at every baseline over Flash3D. Because most auxiliary views are near the source, the pipeline remains dominated by the cheap single-step restorer and only invokes the heavier video generator where it helps (large baselines).

Limitations. The 158-sample subset is selected to contain large-baseline targets and is not the full 3204-frame split; absolute numbers are not directly comparable to full-split reports. The fusion is a fixed distance schedule; we tried a per-pixel content-adaptive alternative (Sec. 4.4) and found it degrades perceptual quality through boundary seams, so a *learned* whole-frame blend that stays seam-free is the more promising extension. Gains, while consistent across metrics and datasets, are modest in magnitude.

Table 8: Stage-2 exploration (8 scenes), all optimized with `lpips_nov`. [15] Liu et al. Deceptive-NeRF/3DGS. ECCV 2024.

Difix variant	LARGE net LPIPS↓	LARGE win LPIPS↓	[16] Liu et al. Infinite Nature: Perpetual View Generation of Natural Scenes from a Single Image. ICCV 2021.
ref_src (source ref)	-0.014	-0.013	
noref (no reference)	+0.003	+0.004	
ref_nn (rendered ref)	+0.005	+0.006	

Table 9: Stage-1 supervision-density exploration (8 scenes). Denser supervision does not help; aux views are highly redundant.

aux-stride	# views	ALL win PSNR	LARGE net LPIPS↓
1 (all)	~24	+0.047	-0.014
2 (half)	~12	+0.046	-0.014
3 (third)	~8	+0.047	-0.013

References

- [1] Szymanowicz et al. Flash3D: Feed-Forward Generalisable 3D Scene Reconstruction from a Single Image. arXiv:2406.04343.
- [2] Wu et al. Difx3D+: Improving 3D Reconstructions with Single-Step Diffusion Models. CVPR 2025.
- [3] Zhang et al. Scene Splatter: Momentum 3D Scene Generation from Single Image with Video Diffusion Model. CVPR 2025.
- [4] Yu et al. ViewCrafter: Taming Video Diffusion Models for High-fidelity Novel View Synthesis. arXiv:2409.02048.
- [5] Li et al. MINE: Towards Continuous Depth MPI with NeRF for Novel View Synthesis. ICCV 2021.
- [6] Zhang et al. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. CVPR 2018.
- [7] Szymanowicz et al. Splatter Image: Ultra-Fast Single-View 3D Reconstruction. CVPR 2024.
- [8] Charatan et al. pixelSplat. CVPR 2024.
- [9] Chen et al. MVSplat. ECCV 2024.
- [10] Xu et al. DepthSplat. CVPR 2025.
- [11] Yang et al. CATSplat. ICCV 2025.
- [12] Liu et al. 3DGS-Enhancer. NeurIPS 2024.
- [13] Wu et al. ReconFusion. CVPR 2024.
- [14] Gao et al. CAT3D. NeurIPS 2024.